Resolume OFX Bridge user guide

OFX Bridge **runs OpenFX plugins (the format DaVinci Resolve uses) inside Resolume, as native FFGL effects**. You point it at a folder of `.ofx.bundle` plugins and it writes one FFGL bundle per plugin into Resolume's effects folder.

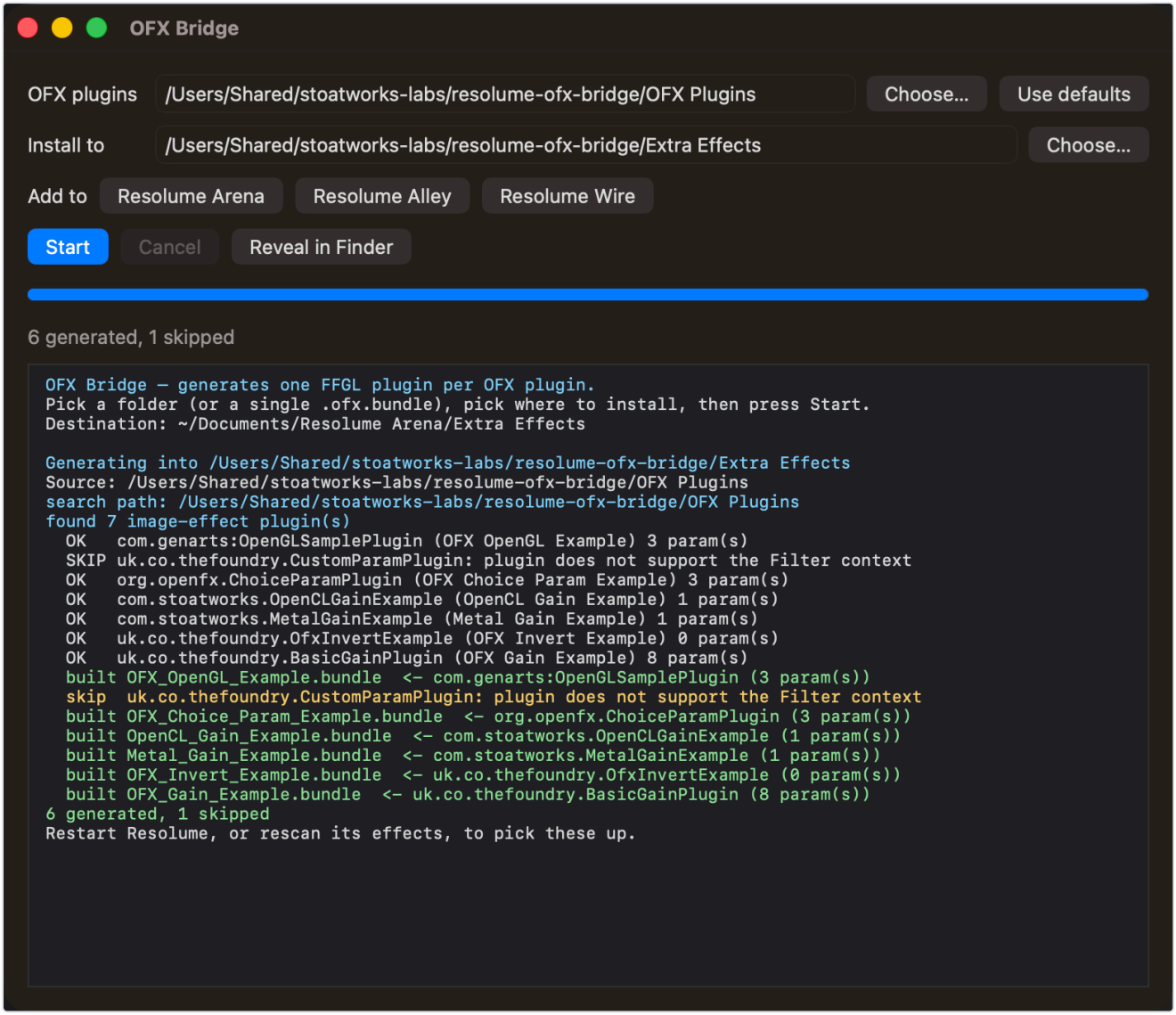
You don't need a compiler. Each generated bundle is a copy of one prebuilt binary plus a JSON manifest that it reads when the host loads it.

> **Before you rely on this:** it's out at v0.10.0 as a working prototype. The host, the parameter mapping and the pixel path are checked against real OFX plugins by automated harnesses, including a headless OpenGL test that drives a generated bundle the way a host would, and generated bundles **have since been run inside Resolume itself on real content**.
>
> There's now a **Windows x64 release as well as the macOS universal build**, and Windows runs the same self-test: it finds the same plugins, declines the same ones, and produces the same pixel values digit for digit. What it *hasn't* done is run the **GL render path** (the one Resolume drives every frame), because that needs a desktop session with an OpenGL 4.1 core driver, and neither the machine here nor a hosted runner is one. That's a narrower gap than "untested", but it's a real one.
>
> **Linux has still never been compiled. Metal and OpenCL are macOS only**: a Windows build advertises neither, so plugins wanting them are declined when they introduce themselves rather than failing part-way through a frame. The **CUDA path is written from the specification and has never been compiled or run**. And **licensed commercial plugins will mostly refuse to load** (see What won't work).

Installing plugins

Download the release, unzip, and open **OFX Bridge.app**: choose a folder of OFX plugins (or a single `.ofx.bundle`), choose where to install, press **Start**.

On Windows there's no app, and none is planned. `ofxgen.exe` does everything the window does, and the command-line section below is the whole story there.



The **Add to** buttons fill in the destination for whichever Resolume products are installed. Only Arena and Avenue are known to scan an `Extra Effects` folder. Buttons appear for Alley and Wire because that's where they *would* look if support arrives, and clicking one says so in the log.

Then **rescan effects in Resolume**.

> The app clears macOS quarantine from what it writes, which matters more than it sounds: **Resolume silently skips a quarantined plugin** rather than asking. If plugins still don't appear after something else has copied them around:
>
> ```
> xattr -dr com.apple.quarantine ~/Documents/Resolume\ Arena/Extra\ Effects
> ```

From the command line

```
./ofxgen generate --out ~/Documents/Resolume\ Arena/Extra\ Effects
```

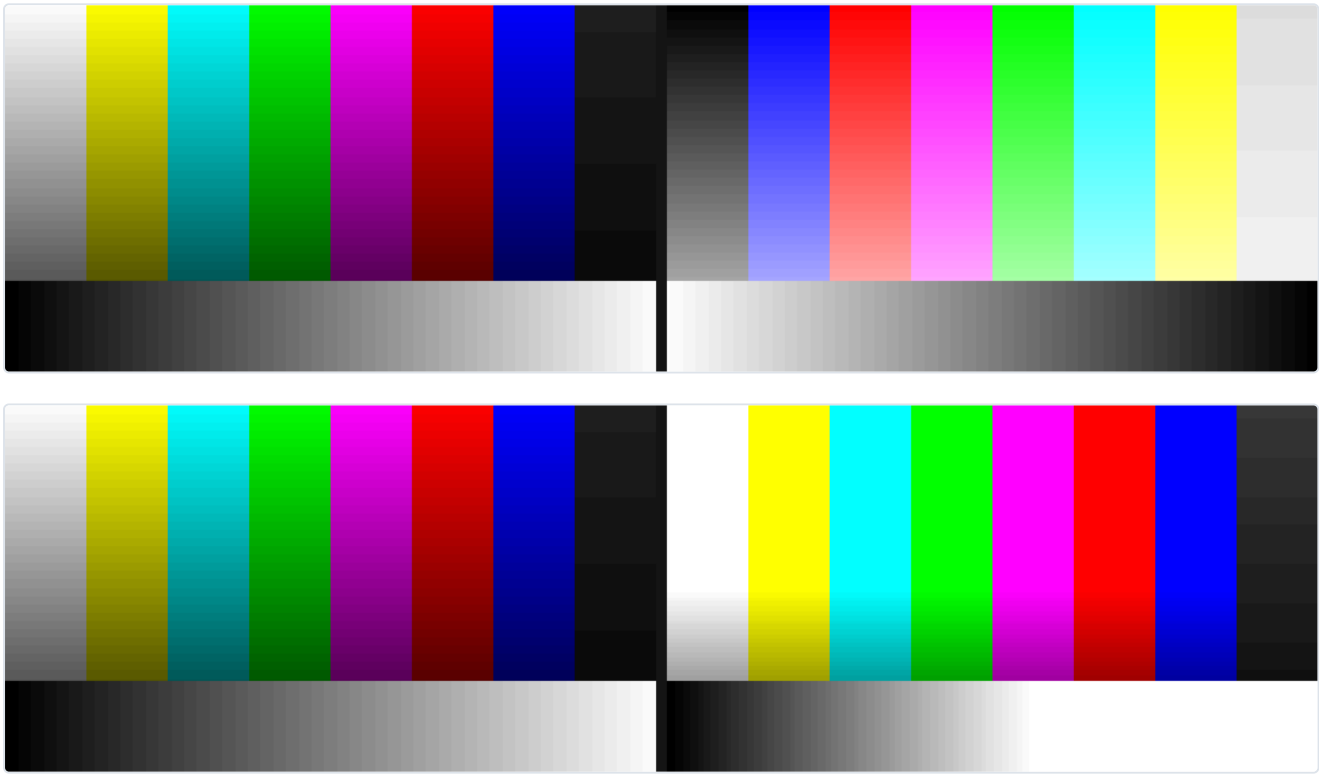
That scans the conventional OFX locations for your platform (plus anything on `OFX_PLUGIN_PATH`). Add `--dir` to scan elsewhere, or `--bundle /path/to/one.ofx.bundle` for a single plugin.

"I have Resolve, so I have OFX plugins" — probably not

There's most likely no OFX plugin on your machine even if you have Resolve. Resolve compiles its own ResolveFX into the application rather than installing loadable bundles, so `/Library/OFX/Plugins` is usually empty.

To try the bridge with nothing installed, build a corpus from the OpenFX examples:

```
./scripts/build-test-plugins.sh
./build/ofxgen generate --dir build/test-plugins --out build/generated
```



Neither image is a mock-up: both are written straight out of the plugin's framebuffer.

Why one bundle per plugin, not a dropdown

The obvious design is a single "OFX Host" effect with a dropdown listing your plugins. **FFGL can't express that.**

FFGL 2.2 can change a parameter's display name, visibility and dropdown elements at runtime, but its header explicitly lists **range and default-value changes as not supported**, and a parameter's

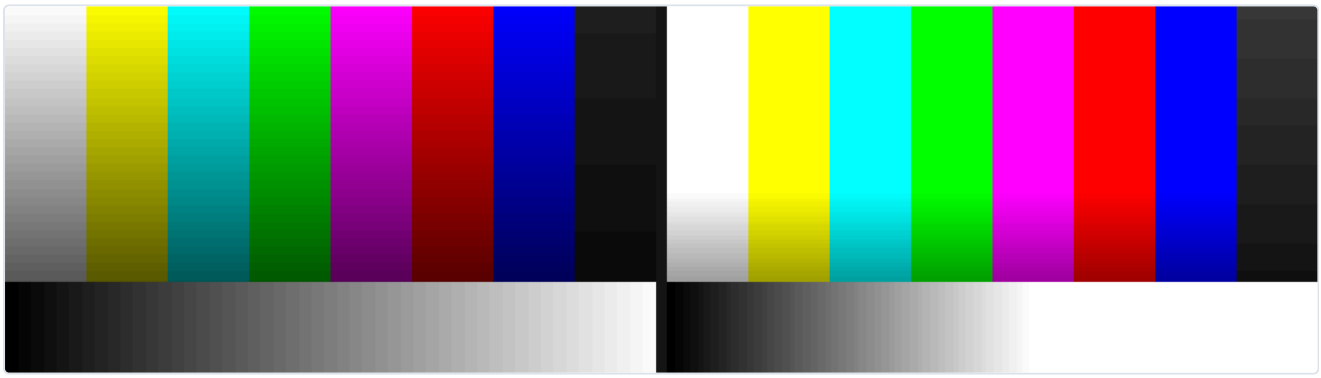
type is fixed when the plugin loads. A dropdown wrapper would have to declare a fixed pool of pre-typed 0–1 sliders and rename them: every parameter would lose its real units, and any plugin with more parameters of a type than the pool has slots couldn't be shown at all.

Generating one bundle per plugin keeps each parameter just as its author declared it: a 0–100 range stays 0–100.

Speed: which path your plugin takes

The plugin advertises	What happens	Speed at 1080p
OFX Metal render	Renders on the GPU, no CPU round trip	**5.4× CPU**
OFX OpenGL render	GPU, supported	—
OpenCL	Implemented and verified	4.2× CPU
Anything else	CPU path — a full GPU → CPU → GPU trip per frame	baseline

Metal is what Resolve-targeted plugins on macOS use, and Metal wins when a plugin offers both Metal and OpenCL.



What won't work

- **Only the Filter context is hosted**, since that's what maps onto an effect slot in a Resolume clip. Generator, Transition and General-only plugins are reported and skipped.
- **Licensed commercial plugins will mostly refuse to load.** The bridge tells plugins the truth about which host it is, and many commercial OFX plugins only license themselves to hosts they recognise. **That's the vendor's decision, and I've chosen not to work around it.**
- **CUDA plugins are declined cleanly.** The path is written but never compiled or run: it needs an NVIDIA GPU, which macOS hasn't supported since 10.13. The host doesn't advertise CUDA, so such plugins are refused rather than half-run.

- **OpenGL-render plugins must use core-profile GL.** Immediate-mode drawing is illegal in a core profile, and macOS has no compatibility profile above 2.1.
- **Parametric (curve) parameters are declined**: FFGL has no equivalent, and any flattening would misrepresent the plugin's UI.

Working out why a plugin didn't appear

That's what the tools are for. Use them in this order:

```
./build/ofxgen list                      # what was found, and why anything was skipped
./build/ofxprobe --dir /path/to/plugins  # a plugin's contexts and parameters
./build/ofxgen verify <bundle>           # load a generated bundle as a host would
./build/ffgltest <bundle>                # drive it through a real OpenGL context
```

`ofxgen list` is the first stop: a plugin that's skipped says so, with a reason.

`ofxgen verify` shows what survived the mapping. For the Gain example, **eight OFX parameters become six FFGL ones**: the group and page carry no value of their own, the group becomes a heading on its children, and the 0–100 ranges survive intact.

The other directions: FFGL and After Effects plugins in Resolve

Since v0.8.0 the bridge also runs the other way. If you have an FFGL effect (one of the Stoatworks set, or any third-party one), `wrap-ffgl` turns it into an OpenFX plugin that Resolve, Vegas, Nuke and Natron load:

```
ofxgen wrap-ffgl --bundle Tinsel.bundle --out /Library/OFX/Plugins
```

And an After Effects plugin can cross too, on the bridge's built-in minimal AE host:

```
ofxgen wrap-ae --bundle "Luma Key.plugin" --out /Library/OFX/Plugins
```

To carry an After Effects plugin all the way into Resolume, run the two generators in sequence: `wrap-ae` first, then `generate` on its output.

Three things to know. Wrapped FFGL effects render at 8 bits per channel (FFGL is an 8-bit world) and one at a time. Effects that keep their own clock will behave oddly when the host scrubs. And the AE host is minimal on purpose: plain CPU effects work, and plugins that need GPU suites, custom UI or a licence check against a running After Effects are refused with the missing piece named in the log rather than guessed around.

Troubleshooting

Symptom	Cause
No plugins found at all	Very likely there are none installed — Resolve doesn't install loadable bundles. Build the test corpus.
A plugin was skipped	Run `ofxgen list` for the reason: wrong context, parametric parameters, or CUDA-only.
Bundle generated but Resolume doesn't show it	Quarantine, almost always — Resolume skips a quarantined plugin silently. Clear it, then rescan effects.
A commercial plugin refuses to license	Expected. It only licenses to hosts it recognises, and the bridge does not spoof one.
Effect renders but is slow	It is on the CPU path — a full GPU round trip per frame. Check whether it advertises Metal.
OpenGL-render plugin draws nothing	It probably uses immediate-mode GL, which a core profile forbids.
A Metal or OpenCL plugin is declined on Windows	Expected — both interops are macOS. The plugin is turned down at describe time rather than failing mid-frame.
Linux build	Never compiled. macOS and Windows x64 only.

See also

- 01-architecture.md: why a generator, and how a copied binary knows which plugin it is
- 02-parameter-mapping.md: how each OFX parameter type becomes FFGL parameters
- 03-verification.md: **what's actually tested, and what isn't**
- 04-gpu-acceleration.md: where a frame actually goes
- README: the tool list and downloads